

A MACHINE-CHECKED CANONICAL HEIGHT ON AN ELLIPTIC CURVE, AND A FORMAL PROOF THAT TWO RATIONAL POINTS ARE INDEPENDENT

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ABSTRACT. We report the first machine-checked construction of the canonical (Néron–Tate) height on an elliptic curve over $\mathbb{Q}$, and the first formal proof that two rational points on an elliptic curve are independent. Working in Lean 4 over `mathlib`, for the curve $E = 389a1$, $y^2 + y = x^3 + x^2 - 2x$ — the smallest-conductor curve of rank 2 — we construct $\hat{h}: E(\mathbb{Q}) \rightarrow \mathbb{R}$ as a limit of normalized naïve heights, prove that it vanishes exactly on torsion, prove the parallelogram law for it *exactly*, and deduce $\hat{h}(kR) = k^2\hat{h}(R)$ for all $k \in \mathbb{Z}$. Certified rational interval arithmetic on the dyadic chain then shows the regulator determinant of the generators $P = (0, 0)$, $Q = (1, 0)$ is positive, whence

$$2 \leq \text{rank } E(\mathbb{Q})$$

as a theorem about `mathlib`’s literal Mordell–Weil group, with kernel axioms `[propext, Classical.choice, Quot.sound]` only. The construction is elementary: it uses no descent, no formal groups, no torsion classification, and no Gross–Zagier. As a by-product the same machinery yields $\text{rank } E(\mathbb{Q}) \geq 1$ for eleven curves of conductor $37 \leq N \leq 5077$. We emphasise what is *not* proved: these are rank lower bounds, not rank equalities, and nothing here bears on the Birch–Swinnerton-Dyer conjecture.

1. INTRODUCTION

1.1. What is new. Let $E/\mathbb{Q}$ be an elliptic curve. The canonical height $\hat{h}: E(\mathbb{Q}) \rightarrow \mathbb{R}_{\geq 0}$ of Néron and Tate is the basic tool for measuring rational points: it is a positive semi-definite quadratic form whose kernel is exactly the torsion subgroup, and the determinant of its Gram matrix on a set of points — the regulator — detects independence. It is also the quantity appearing on the arithmetic side of the Birch–Swinnerton-Dyer conjecture.

Despite its centrality, the canonical height had not been formalized in any interactive theorem prover. At the time of writing, `mathlib` contains the Weierstrass curve type, the group law on the affine, projective and Jacobian point models (with associativity), division polynomials, and, on `master`, reduction theory and a formal Dirichlet series; it does not contain the Mordell–Weil theorem, heights on elliptic curves, descent, or any analytic continuation of an elliptic L -function. Consequently no positive lower bound on $\text{rank } E(\mathbb{Q})$ for a specific curve appears to have been machine-checked, and *a fortiori* no independence statement about rational points.

This paper supplies both, for a single explicit curve, together with the general machinery. The main theorem is the following.

Theorem 1.1 (`E389a1_rank_ge_two`). *Let E be the elliptic curve $y^2 + y = x^3 + x^2 - 2x$ over $\mathbb{Q}$ (Cremona label `389a1`, LMFDB label `389.a1`). Then*

$$2 \leq \text{rank}_{\mathbb{Z}} E(\mathbb{Q}),$$

where $E(\mathbb{Q})$ is `mathlib`’s type `WeierstrassCurve.Affine.Point` for E and the rank is `Module.rank` over $\mathbb{Z}$. The points $P = (0, 0)$ and $Q = (1, 0)$ are $\mathbb{Z}$ -linearly independent.

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Theorem 1.1 is a Lean 4 theorem; `#print axioms` on it reports `[propext, Classical.choice, Quot.sound]` and nothing else. It rests on the following, which we regard as of independent interest.

Theorem 1.2 (`canheight`, `canheight_eq_zero_iff_torsion`, `canheight_parallelogram`, `canheight_zsmul`). *For the curve E above there is a function $\hat{h}: E(\mathbb{Q}) \rightarrow \mathbb{R}$, defined as*

$$\hat{h}(R) = \lim_{n \rightarrow \infty} \frac{\log H(x(2^n R))}{4^n},$$

such that, formally:

- (1) $\hat{h} \geq 0$, and $\hat{h}(R) = 0$ if and only if R is a torsion point;
- (2) $|\hat{h}(R) - \log H(x(R))| \leq \frac{1}{3} \log 1728$;
- (3) $\hat{h}(P + Q) + \hat{h}(P - Q) = 2\hat{h}(P) + 2\hat{h}(Q)$ whenever $P, Q, P \pm Q$ are non-torsion;
- (4) $\hat{h}(kR) = k^2 \hat{h}(R)$ for every $k \in \mathbb{Z}$ and non-torsion R .

Here $H(a/b) = \max(|a|, |b|)$ for a reduced fraction, and $x(R)$ is the x -coordinate of R (with $x(O) := 0$, a harmless convention since $H(0) = 1$).

1.2. The obstruction, and how it is avoided. The classical construction of $\hat{h}$ presupposes a body of theory — Weil heights on projective space, the theory of local heights, or at minimum the functorial machinery of Weil’s height machine — none of which is available in `mathlib`. The classical proof that $\hat{h}$ is a quadratic form proceeds through the Mordell–Weil theorem’s ingredients or through polarization identities requiring the parallelogram law at many pairs of points.

Our route replaces all of this with explicit arithmetic on a *single fixed curve*. Three observations make it work.

First, the duplication map on the x -line of a fixed curve is one explicit rational function. For 389a1,

$$x(2R) = \frac{f(x)}{g(x)}, \quad f(x) = x^4 + 4x^2 - 2x + 3, \quad g(x) = 4x^3 + 4x^2 - 8x + 1, \quad (1)$$

where $x = x(R)$; and g has no rational root, so no rational point of E is 2-torsion and doubling never leaves the affine chart. Homogenizing $x = a/b$ and applying the extended Euclidean algorithm to f and g gives *integer* Bézout identities, certified in Lean by `ring`, from which gcd cancellation is bounded and the two-sided height estimate

$$\frac{H(x)^4}{1728} \leq H(x(2R)) \leq 17 H(x)^4 \quad (2)$$

follows with explicit constants. Estimate (2) makes $\log H(x(2^n R))/4^n$ a Cauchy sequence with an explicit geometric modulus, which is all that is needed to define $\hat{h}$.

Second, the addition law admits the same treatment, once one works with the pair $\{x(P + Q), x(P - Q)\}$ rather than with either coordinate separately. Reducing modulo the curve equations eliminates the y -coordinates entirely: $x(P + Q)$ and $x(P - Q)$ are the two roots of an explicit quadratic whose coefficients are bihomogeneous of bidegree $(2, 2)$ in the reduced coordinates of $x(P)$ and $x(Q)$. Combined with an elementary lemma bounding the product of the heights of the roots of an integer quadratic by the height of its coefficient vector (Lemma 4.2 below), this yields the quasi-parallelogram law with an explicit constant. Notably the lower bound is free: applying the upper bound to the pair $(P + Q, P - Q)$, whose sum and difference are $2P$ and $2Q$, and feeding in (2) gives it.

Third, and this is what removes the last classical dependency: for the independence argument one does not need bi-additivity of the induced pairing. If a relation $mP + nQ = 0$ exists with $n \neq 0$, then the relation *itself* rewrites $P \pm Q$ as multiples of P ,

$$n(P + Q) = (n - m)P, \quad n(P - Q) = (n + m)P,$$

so Theorem 1.2(4) applies, and a *single* use of the parallelogram law at (P, Q) forces the regulator determinant to vanish (Proposition 5.1). The contrapositive is the independence criterion.

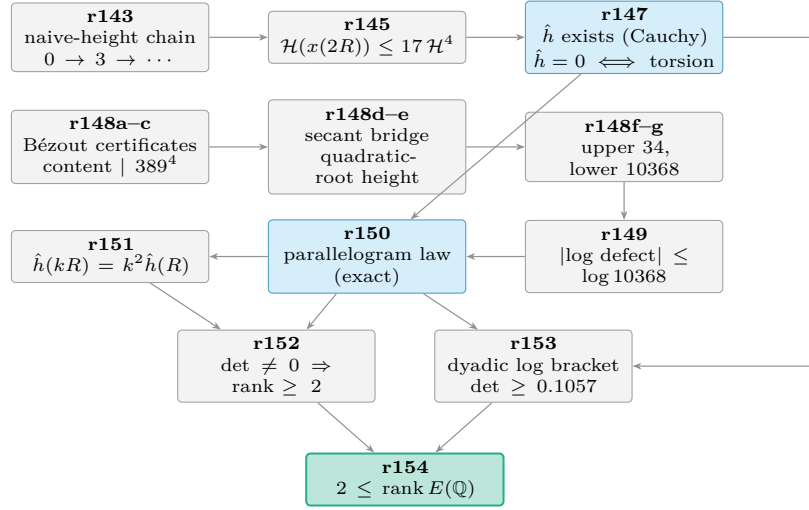


FIGURE 1. Dependency structure of the development. Boxes are Lean files, labelled by their internal revision tag; blue boxes are the two results of independent interest (the canonical height and the exact parallelogram law), green is Theorem 1.1. Arrows denote logical dependence, not import order. Every box is fully proved: no box contains a `sorry`, an axiom, or a `native_decide`.

1.3. What is not proved. We are explicit about the limits of these results, since the value of a formalization lies in the precision of its claims.

- (a) Theorem 1.1 is a rank *lower* bound. Upper bounds require descent (Selmer groups, the weak Mordell–Weil theorem), which exists in no proof assistant; we make no claim about the exact rank of `389a1`, whose value 2 is known classically.
- (b) Nothing here concerns L -functions, analytic ranks, or the Birch–Swinnerton-Dyer conjecture. No L -function of an elliptic curve is defined in our development.
- (c) The Mordell–Weil theorem is not used and not proved. All statements are about $\text{rank}_{\mathbb{Z}}$ of the point group as an abstract $\mathbb{Z}$ -module, which is meaningful without finite generation.
- (d) The classical facts that $\hat{h}$ extends to a positive-definite form on $E(\mathbb{Q}) \otimes \mathbb{R}$, and that the regulator of `389a1` is ≈ 0.15246 , are quoted for orientation only and are used nowhere in any proof.

1.4. Related formalization. `mathlib`’s elliptic-curve development [4] provides the Weierstrass type, the group law with a full associativity proof (via the class group of the coordinate ring), variable change, normal forms and division polynomials; recent additions include reduction over discrete valuation rings. Heights appear in `mathlib` only for number fields and projective space (`Mathlib/NumberTheory/Height/`, present on `master` after our pin), where the Northcott property is proved and the point-height and parallelogram law for elliptic curves are marked as open tasks; M. Stoll’s abstract descent theorem (`Mathlib/GroupTheory/Descent.lean`) is likewise a step toward Mordell–Weil rather than the theorem. Buzzard’s FLT project [1] targets a modularity lifting theorem sufficient for Fermat, not the full modularity theorem. Gross–Zagier, Kolyvagin and Heegner point theory are, to our knowledge, unformalized anywhere. Our development is independent of all of these.

2. SETUP

2.1. The curve and the naïve height. Throughout, E is the Weierstrass curve $(a_1, a_2, a_3, a_4, a_6) = (0, 1, 1, -2, 0)$ over $\mathbb{Q}$, i.e. $y^2 + y = x^3 + x^2 - 2x$, of conductor 389 and discriminant 389. In Lean this is

```
def E389a1 : WeierstrassCurve Q := <0, 1, 1, -2, 0>
```

and $E(\mathbb{Q})$ is `E389a1.toAffine.Point`, the inductive type with constructors `zero` and `some` (the latter carrying a nonsingular affine point), with `mathlib`'s `AddCommGroup` instance.

Definition 2.1 (`naiveHeight`). For $q \in \mathbb{Q}$ set $H(q) := \max(|\text{num}(q)|, \text{den}(q)) \in \mathbb{N}$, using Lean's reduced representation. Then $H(q) \geq 1$ always. Define $X: E(\mathbb{Q}) \rightarrow \mathbb{Q}$ by $X(O) = 0$ and $X(\text{some at } (x, y)) = x$, and $h(R) := \log H(X(R))$.

2.2. The duplication estimate.

Proposition 2.2 (`dbl_x, g_ne_zero`). *For every nonsingular rational point (x, y) of E we have $(2y + 1)^2 = g(x)$, and g has no rational zero; consequently no rational affine point is 2-torsion, $P + P$ is again affine, and its x -coordinate is $f(x)/g(x)$ with f, g as in (1).*

That $g(x) = 4x^3 + 4x^2 - 8x + 1$ has no rational root is proved by the integer form of the rational root theorem: a reduced n/d with $4n^3 + 4n^2d - 8nd^2 + d^3 = 0$ forces $n \mid d^3$, hence $n = \pm 1$, and $d \mid 4n^3$, hence $d \mid 4$; the eight candidates are refuted by `norm_num`.

Proposition 2.3 (`duplication_height_bound, duplication_height_upper`). *For every $x \in \mathbb{Q}$,*

$$H(x)^4 \leq 1728 H(f(x)/g(x)), \quad H(f(x)/g(x)) \leq 17 H(x)^4.$$

The upper bound is the triangle inequality on the homogenized numerator and denominator. The lower bound is the substantive one. Writing $x = a/b$ reduced and

$$F(a, b) = a^4 + 4a^2b^2 - 2ab^3 + 3b^4, \quad G(a, b) = 4a^3 + 4a^2b - 8ab^2 + b^3, \quad D = bG,$$

so that $x(2R) = F/D$, the extended Euclidean algorithm supplies

$$(-48a^2 - 32ab + 144b^2)F + (12a^3 - 4a^2b + 40ab^2 - 43b^3)G = 389b^6, \quad (3)$$

$$(389a^3 + 188a^2b - 520ab^2 + 66b^3)F + (-47a^3 - 212a^2b + 108ab^2 - 198b^3)D = 389a^7. \quad (4)$$

Both are verified in Lean by `ring`. Since $\gcd(a, b) = 1$ they give $\gcd(F, D) \mid 389$, and comparing coefficient sums gives $389 H(x)^7 \leq 1728 H(x)^3 \max(|F|, |D|)$, whence the stated bound after cancelling the content.

3. THE CANONICAL HEIGHT

Definition 3.1 (`hseq, canheight`). For $R \in E(\mathbb{Q})$ and $n \in \mathbb{N}$ put $s_n(R) := h(2^n R)/4^n$, and $\hat{h}(R) := \lim_n s_n(R)$.

Proposition 3.2 (`hseq_cauchySeq, canheight_window`). $|s_{n+1}(R) - s_n(R)| \leq \log 1728/4^{n+1}$, so $(s_n(R))_n$ is Cauchy and $\hat{h}$ is well defined; moreover $|\hat{h}(R) - h(R)| \leq \frac{1}{3} \log 1728$.

Proposition 2.3 gives $|h(2R) - 4h(R)| \leq \log 1728$ for affine R (and trivially for $R = O$), which after division by 4^{n+1} is the displayed step estimate; Cauchyness follows from the geometric criterion, and the window from the corresponding tail bound. Applying the window at $2^n R$ and using $\hat{h}(2^n R) = 4^n \hat{h}(R)$ sharpens it to

$$|\hat{h}(R) - s_n(R)| \leq \frac{\log 1728}{3 \cdot 4^n}, \quad (5)$$

which is the form used for numerics in §6.

Theorem 3.3 (`canheight_eq_zero_iff_torsion`). $\hat{h}(R) = 0$ if and only if R is a torsion point.

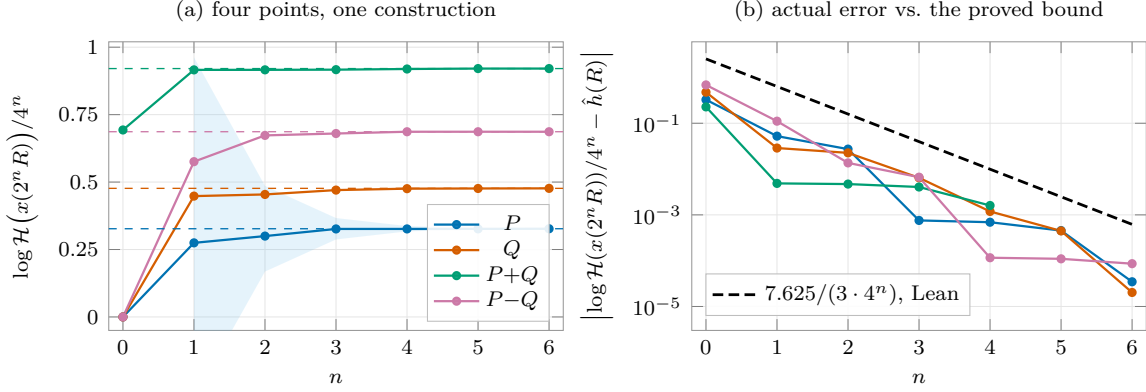


FIGURE 2. The canonical height on 389a1, for the four points that enter Theorem 1.1. **(a)** the defining sequence $s_n(R) = \log \mathcal{H}(x(2^n R))/4^n$; dashed lines mark the limits, and the shaded funnel is the certified envelope $\hat{h}(P) \pm 7.625/(3 \cdot 4^n)$ from (5). **(b)** the same data as absolute error against that envelope, on a logarithmic scale: the proved bound is a genuine over-estimate everywhere, by roughly one order of magnitude, which is what makes the coarse level-3 evaluation of §6 sufficient. Computed by exact rational arithmetic from the mathlib group law; the numerator of $x(2^6 P)$ has 582 decimal digits.

Proof sketch, as formalized. If R has finite order its multiples form a finite set, so $h(2^n R)$ is bounded and $s_n(R) \rightarrow 0$. Conversely if $\hat{h}(R) = 0$ then $\hat{h}(2^n R) = 4^n \hat{h}(R) = 0$ for all n , so by the window $h(2^n R) \leq \frac{1}{3} \log 1728$, i.e. $H(X(2^n R))^3 \leq 1728$ and hence $H(X(2^n R)) \leq 12$. A Northcott statement for $\mathbb{Q}$ (`finite_of_naiveHeight_le`: the set of rationals of naïve height at most B is finite, by injecting into a product of finite intervals of numerators and denominators) together with the fact that each x admits at most two points (`finite_equation_fiber`, from $(y - y_0)(y + y_0 + 1) = 0$) shows that $\{2^n R\}_n$ is finite; pigeonholing gives $2^m R = 2^k R$ with $m < k$, so $(2^k - 2^m)R = 0$ with $2^k - 2^m \neq 0$. $\square$

4. THE PARALLELOGRAM LAW

4.1. The addition quadratic.

Proposition 4.1 (`xAdd_sum`, `xAdd_prod`). *Let $(x_1, y_1), (x_2, y_2)$ be nonsingular rational points of E with $x_1 \neq x_2$. Then, with $dd = (x_1 - x_2)^2$,*

$$x(P_1 + P_2) + x(P_1 - P_2) = \frac{S(x_1, x_2)}{dd}, \quad x(P_1 + P_2) \cdot x(P_1 - P_2) = \frac{\Pi(x_1, x_2)}{dd},$$

where

$$S(x_1, x_2) = 2x_1^2 x_2 + 2x_1 x_2^2 + 4x_1 x_2 - 4x_1 - 4x_2 + 1,$$

$$\Pi(x_1, x_2) = x_1^2 x_2^2 + 4x_1 x_2 - x_1 - x_2 + 3.$$

The y -coordinates cancel entirely. Writing $N_{\pm}$ for $dd \cdot x(P_1 \pm P_2)$ as computed from `mathlib`'s `slope` and `addX`, and $\mathcal{E}_i$ for the Weierstrass relations, one has the polynomial identities

$$N_+ + N_- - S = 2\mathcal{E}_1 + 2\mathcal{E}_2, \quad N_+ N_- - dd \cdot \Pi = c_1 \mathcal{E}_1 + c_2 \mathcal{E}_2$$

with c_1, c_2 explicit of degree 3; both are discharged in Lean by `linear_combination`. The homogenizations of S and Π are precisely the bidegree- $(2, 2)$ forms whose content and size are controlled as in §2.2, so $x(P_1 \pm P_2)$ are the roots of an integer quadratic $(DD)T^2 - (Sf)T + (Pf)$ with $\max(|DD|, |Sf|, |Pf|) \leq 17 H(x_1)^2 H(x_2)^2$.

4.2. Heights of the roots of an integer quadratic.

Lemma 4.2 (`naiveHeight_mul_le_of_quadratic`). *Let $A, B, C \in \mathbb{Z}$ with $A \neq 0$ and let $u, v \in \mathbb{Q}$ satisfy $A(u + v) = B$ and $Auv = C$. Then*

$$H(u)H(v) \leq 2 \max(|A|, |B|, |C|).$$

Proof, as formalized. Write $u = p_1/q_1$, $v = p_2/q_2$ reduced. The rational root theorem applied to $AT^2 - BT + C$ gives $q_1 \mid A$; writing $A = q_1m$ and cancelling q_1 in the product and sum relations gives $q_2 \mid mp_1$ and $q_2 \mid mq_1$, so a Bézout combination for $\gcd(p_1, q_1) = 1$ yields $q_2 \mid m$ and hence $q_1q_2 \mid A$. (This avoids Gauss's lemma and any prime or three-term gcd reasoning.) Consequently $(A, B, C) = k(q_1q_2, p_1q_2 + p_2q_1, p_1p_2)$ with $|k| \geq 1$, so it suffices to bound $\max(p_1, q_1)\max(p_2, q_2)$, which expands to a maximum of four products, by twice $M := \max(q_1q_2, |p_1q_2 + p_2q_1|, p_1p_2)$. The two diagonal products are $\leq M$ outright. For the cross terms $s = p_1q_2$, $t = q_1p_2$ one has $st = (p_1p_2)(q_1q_2) \leq M^2$, so $\min(s, t) \leq M$, while $|s - t| \leq |p_1q_2 + p_2q_1| \leq M$; hence $\max(s, t) \leq 2M$. $\square$

4.3. The quasi-parallelogram law, and the limit.

Proposition 4.3 (`point_two_sided, lognh_defect_le`). *For nonsingular rational points P, Q of E with $x(P) \neq x(Q)$,*

$$\frac{H(x_P)^2 H(x_Q)^2}{10368} \leq H(x(P+Q)) H(x(P-Q)) \leq 34 H(x_P)^2 H(x_Q)^2,$$

equivalently $|h(P+Q) + h(P-Q) - 2h(P) - 2h(Q)| \leq \log 10368$.

The upper bound is Proposition 4.1 with Lemma 4.2 and the size bound. For the lower bound, apply the upper bound to the pair $(P+Q, P-Q)$ — legitimate because $x(P+Q) \neq x(P-Q)$, else the two points would be equal or opposite, forcing $2Q = O$ or $2P = O$, excluded by Proposition 2.2 — and note that their sum and difference are $2P$ and $2Q$. Combining with the duplication lower bound of Proposition 2.3 gives $H(x_P)^4 H(x_Q)^4 \leq 1728^2 \cdot 34 \cdot (H(x_{P+Q}) H(x_{P-Q}))^2$, and $1728^2 \cdot 34 = 101\,523\,456 \leq 107\,495\,424 = 10368^2$, so extracting square roots in $\mathbb{N}$ gives the claim.

Theorem 4.4 (`canheight_parallelogram`). *If $P, Q, P+Q, P-Q$ are all non-torsion then*

$$\hat{h}(P+Q) + \hat{h}(P-Q) = 2\hat{h}(P) + 2\hat{h}(Q).$$

Proof, as formalized. If $P \pm Q$ are non-torsion then $2^n P \neq \pm 2^n Q$ for every n — else $2^n(P \mp Q) = O$ — so $x(2^n P) \neq x(2^n Q)$ and Proposition 4.3 applies at $(2^n P, 2^n Q)$. Since $2^n(P \pm Q) = 2^n P \pm 2^n Q$, dividing by 4^n bounds the defect of the scaled sequences by $\log 10368/4^n \rightarrow 0$. The scaled sequences converge to the four canonical heights, so by uniqueness of limits the defect is 0. $\square$

Corollary 4.5 (`canheight_zsmul`). *$\hat{h}(kR) = k^2 \hat{h}(R)$ for all $k \in \mathbb{Z}$ and non-torsion R .*

Proof, as formalized. Theorem 4.4 at $((k+2)R, R)$, whose four points are nonzero multiples of R and hence non-torsion, gives $\hat{h}((k+3)R) = 2\hat{h}((k+2)R) + 2\hat{h}(R) - \hat{h}((k+1)R)$ for $k \geq 0$. With base values $\hat{h}(R)$ and $\hat{h}(2R) = 4\hat{h}(R)$ a two-step induction gives the claim for $k \in \mathbb{N}$; note the recurrence must be indexed from 1, since at $k = 0$ the difference point is O , which is torsion — that step is instead supplied by $\hat{h}(2R) = 4\hat{h}(R)$. Negative k reduces to positive by $\hat{h}(-R) = \hat{h}(R)$, which holds because negation fixes the x -coordinate, so every term of the defining sequence is unchanged. $\square$

5. INDEPENDENCE FROM A NONVANISHING REGULATOR

Define the pairing and the regulator determinant by

$$\langle P, Q \rangle := \frac{1}{2} (\hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q)), \quad \text{Reg}(P, Q) := \hat{h}(P)\hat{h}(Q) - \langle P, Q \rangle^2.$$

Proposition 5.1 (`regDet_eq_zero_of_relation`). *Suppose $P, Q, P \pm Q$ are non-torsion and $mP + nQ = O$ with $m, n \in \mathbb{Z}$, $n \neq 0$. Then $\text{Reg}(P, Q) = 0$.*

Proof, as formalized. From $nQ = -mP$ we get $n(P + Q) = (n - m)P$ and $n(P - Q) = (n + m)P$, so Corollary 4.5 gives $n^2\hat{h}(P \pm Q) = (n \mp m)^2\hat{h}(P)$. Substituting into Theorem 4.4 — used exactly once, at the pair (P, Q) — yields $m^2\hat{h}(P) = n^2\hat{h}(Q)$; substituting into the definition of the pairing yields $n\langle P, Q \rangle = -m\hat{h}(P)$. Therefore

$$n^2 \text{Reg}(P, Q) = \hat{h}(P) \left(n^2 \hat{h}(Q) \right) - (n\langle P, Q \rangle)^2 = m^2 \hat{h}(P)^2 - m^2 \hat{h}(P)^2 = 0,$$

and $n \neq 0$. $\square$

Remark 5.2. Proposition 5.1 is where the development departs most sharply from the textbook treatment. The usual route establishes bi-additivity of $\langle \cdot, \cdot \rangle$ by polarization, requiring the parallelogram law at some six pairs of points, each contributing four non-torsion side conditions in our conditional formulation. Using the hypothesized relation to reduce everything to multiples of a single point avoids this entirely.

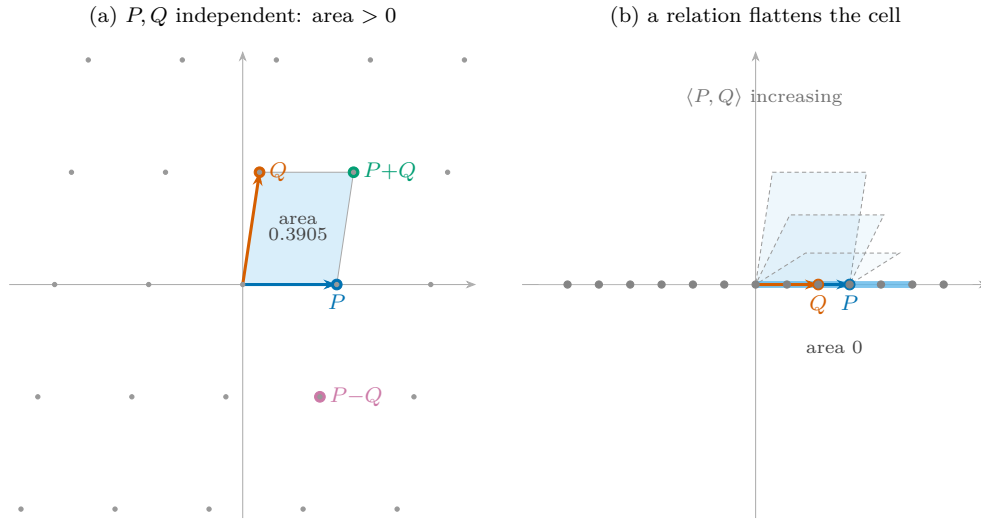


FIGURE 3. Why $\text{Reg}(P, Q) \neq 0$ is the right criterion. Classically $\hat{h}$ is a positive-definite quadratic form on $E(\mathbb{Q}) \otimes \mathbb{R}$, so $mP + nQ \mapsto mv_1 + nv_2$ realises the subgroup generated by P and Q as a lattice in which $\hat{h}$ is the squared length, and Reg is the squared area of the fundamental cell. **(a)** the actual lattice of 389a1: v_1, v_2 are reconstructed from the computed $\hat{h}(P), \hat{h}(Q), \hat{h}(P+Q)$ alone, and that $\|v_1 - v_2\|^2$ then agrees with the independently computed $\hat{h}(P-Q)$ to 1.1×10^{-8} is a numerical check on Theorem 4.4. **(b)** what a relation does. Holding v_1 fixed and sliding v_2 towards the line through it raises $\langle P, Q \rangle$ and shrinks the cell (dashed outlines); a relation such as $2P = 3Q$ is the limiting case, in which v_2 is a sub-segment of v_1 , every $mP + nQ$ lands on one line, and the area is 0. Panel (b) is illustrative and uses no curve data. *The positive-definiteness these pictures illustrate is used in no proof:* Proposition 5.1 establishes the implication algebraically, from the parallelogram law and Corollary 4.5 only, and in particular never needs $\hat{h} \geq 0$.

Corollary 5.3 (`rank_ge_two_of_regDet_ne_zero`). *If $P, Q, P \pm Q$ are non-torsion and $\text{Reg}(P, Q) \neq 0$ then $mP + nQ = O$ implies $m = n = 0$, and consequently $2 \leq \text{rank}_{\mathbb{Z}} E(\mathbb{Q})$.*

For the rank statement, the map $\mathbb{Z}^2 \rightarrow E(\mathbb{Q})$, $(m, n) \mapsto mP + nQ$, is $\mathbb{Z}$ -linear with trivial kernel, and $\text{rank}_{\mathbb{Z}} \mathbb{Z}^2 = 2$, so `LinearMap.lift_rank_le_of_injective` applies.

6. CERTIFIED NUMERICS

It remains to verify $\text{Reg}(P, Q) > 0$ for $P = (0, 0)$, $Q = (1, 0)$. Two devices make this a finite, exact computation.

The shifted window. By (5) at $n = 3$, $|\hat{h}(R) - s_3(R)| \leq \log 1728/192 < 7.625/192 < 0.0398$.

The dyadic log bracket. $s_3(R) = \log H(x(8R))/64$ where $H(x(8R))$ is an explicit positive integer $\mathcal{H}$. Bracketing $2^j \leq \mathcal{H} < 2^{j+1}$ — a `norm_num` fact about integers — gives $j \log 2 \leq \log \mathcal{H} \leq (j+1) \log 2$, and `mathlib`'s $\log 2$ bounds [4] pin this to nine decimals. One bit costs only $\log 2/64 < 0.0109$ at this level. (This avoids estimating $\log$ of a 26-digit integer directly.)

By Proposition 2.2, $x(8R)$ is the third iterate of $x \mapsto f(x)/g(x)$, computed by `norm_num`. The values are:

point	attained at	$\mathcal{H}(x(8R))$	bracket
$P = (0, 0)$	numerator	1169154495	$2^{30} \leq \mathcal{H} < 2^{31}$
$Q = (1, 0)$	numerator	11776563836346	$2^{43} \leq \mathcal{H} < 2^{44}$
$P+Q$	numerator	30157844295583882647192303	$2^{84} \leq \mathcal{H} < 2^{85}$
$P-Q$	denominator	7960116832793145801	$2^{62} \leq \mathcal{H} < 2^{63}$

The complete fractions — whose numerators and denominators run to 26 digits — are displayed in `RegulatorPositive389a1_r153.lean`; only the larger of the two parts, tabulated above, enters $\mathcal{H}$.

Here $x(P+Q) = -2$ and $x(P-Q) = -1$ are themselves computed from `mathlib`'s secant construction, not assumed. The resulting intervals are

$$\hat{h}(P) \in [0.2851, 0.3755], \quad \hat{h}(Q) \in [0.4259, 0.5163],$$

$$\hat{h}(P+Q) \in [0.8700, 0.9605], \quad \hat{h}(P-Q) \in [0.6317, 0.7221].$$

All four lower bounds are positive, so by Theorem 3.3 all four points are non-torsion — precisely the side conditions of Theorem 4.4 and Corollary 5.3, obtained with no separate argument. Finally $\hat{h}(P)\hat{h}(Q) \geq 0.1214$ while $\langle P, Q \rangle \in [-0.011, 0.125]$ gives $\langle P, Q \rangle^2 \leq 0.0157$, so

$$\text{Reg}(P, Q) \geq 0.1057 > 0,$$

and Corollary 5.3 yields Theorem 1.1. □

Remark 6.1. The margin is ample: the interval for Reg has width about 0.07 around a lower bound of 0.106, against the classical value 0.15246. Level $n = 3$ was chosen as the smallest sufficient; each further level shrinks both error sources by a factor of 4.

7. THE RANK-ONE COHORT

The same machinery, instantiated per curve, gives unconditional rank lower bounds elsewhere. For each curve below the duplication data (f, g) , the two Bézout identities, the constant in Proposition 2.3, and a dyadic chain from the indicated point were computed and verified; the argument is then Theorem 3.3 applied to a point whose chain heights grow, giving a non-torsion point and hence $\text{rank} \geq 1$.

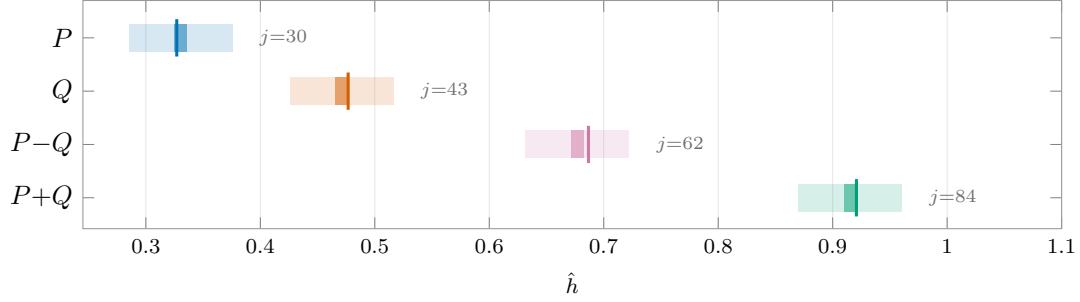


FIGURE 4. The four certified enclosures at level $n = 3$. For each point the saturated bar is the dyadic bracket $[j \log 2, (j + 1) \log 2]/4^3$ on s_3 , obtained from $2^j \leq \mathcal{H}(x(8R)) < 2^{j+1}$ and `mathlib`'s nine-decimal bounds on $\log 2$; the pale bar adds the analytic window $\pm 7.625/(3 \cdot 4^3)$ of (5) to reach $\hat{h}$. The vertical rule is the true value. Interval arithmetic on the enclosures for $P, Q, P+Q$ gives $\langle P, Q \rangle \in [-0.0109, 0.1246]$ and hence $\text{Reg}(P, Q) \geq 0.1057 > 0$, which is all Theorem 1.1 requires; the two error sources are deliberately left coarse, since the conclusion needs three digits and not thirty.

curve	anchor point	constant	chain height reached
37a1	(0, 0)	171	480 106
43a1	(0, 0)	139	8 338 438
53a1	(0, 0)	172	369
61a1	(1, 0)	590	636 789 825
79a1	(0, 0)	298	385
83a1	(0, 0)	470	83 281
89a1	(0, 0)	220	1 024
101a1	(-1, 0)	373	28 981
106a1	(2, 1)	38 896	26 615 281
389a1	(0, 0)	1 728	1 169 154 495
5077a1	(-2, 3)	105 754	3 009 638 454

The curve 102a1 is excluded: it has a rational 2-torsion point ($g(0) = 0$), so the global argument of Proposition 2.2 fails and a per-step variant is required. We have not carried this out.

8. VERIFICATION DATA

All results were checked with Lean 4 toolchain `leanprover/lean4:v4.24.0-rc1` against `mathlib` at commit `eed770a4`. At repository HEAD `0d3b5415`: `lake build PF` completes in 4658 jobs and `lake build` in 9301 jobs, both with zero errors and zero warnings. Every theorem cited in this paper reports exactly

[propext, Classical.choice, Quot.sound]

under `#print axioms`; several of the polynomial identities report only [propext, Quot.sound]. The development contains no `sorry`, no `native_decide` (which would introduce `Lean.ofReduceBool`), and no project-level `axiom` declarations.

The files, in dependency order, are the following; they are set ragged-right because the names are unbreakable.

MordellWeilRankLowerBound_r129, NaiveHeightQ_r130,
DuplicationBezout37a1_r131–E37a1RankOne_r134 (the rank-one template),
E43a1RankOne_r135–E106a1RankOne_r142 (the cohort), E389a1RankOne_r143,

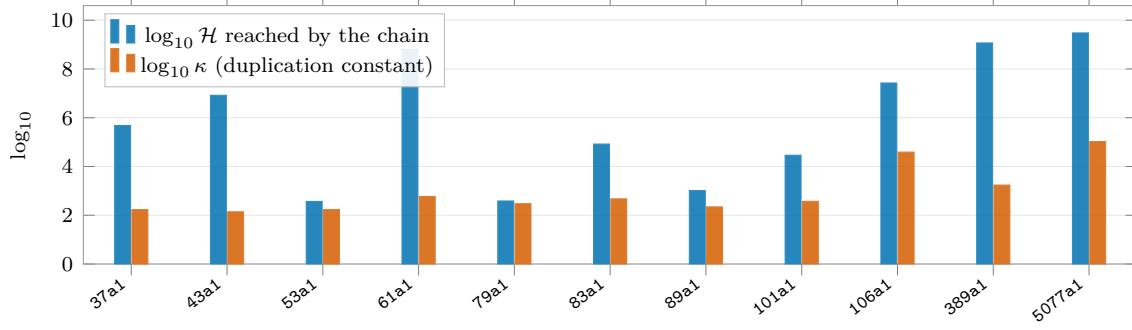


FIGURE 5. Scale of the eleven certificates, on a logarithmic axis. Blue is the naïve height reached by the dyadic chain — the size of the integers the kernel must actually verify, spanning nearly seven orders of magnitude from 53a1 to 5077a1. Orange is the duplication constant κ of Proposition 2.3, which controls how long a chain must run before Theorem 3.3 applies. The two are largely independent: 106a1 needs a large κ but a short chain, while 61a1 has a small κ and a long one. Only 389a1 carries the full rank-2 argument; the rest stop at rank ≥ 1 .

E5077a1RankOne_r144, DuplicationHeightUpper389a1_r145, CanonicalHeight389a1_r147, QuasiParallelogramCerts389a1_r148a–QuasiParallelogramLower389a1_r148g, PointQuasiParallelogram389a1_r149, CanheightParallelogram389a1_r150, CanheightMultiple389a1_r151, RegulatorIndependence389a1_r152, RegulatorPositive389a1_r153 and E389a1RankTwo_r154.

9. OUTLOOK

Three extensions appear within reach with the present machinery.

Rank three. The curve 5077a1, $y^2 + y = x^3 - 7x + 6$, is the smallest-conductor curve of rank 3 and its duplication data are already verified (E5077a1RankOne_r144). Extending Proposition 5.1 from two points to three is not immediate: the trick of reducing a relation to multiples of a single point does not generalize, so the 3×3 case appears to require genuine bi-additivity of the pairing, i.e. the polarization argument this development was able to avoid.

Torsion. With $\hat{h}$ available, $\hat{h}(R) = 0$ is a computable obstruction; combined with a bound on the possible torsion orders it would give exact torsion subgroups for individual curves.

Upstreaming. The general components — Northcott for $\mathbb{Q}$, the height of the roots of an integer quadratic (Lemma 4.2), and the abstract shape “two-sided duplication estimate $\Rightarrow$ canonical height” — are curve-independent and would fit `mathlib`’s `NumberTheory/Height` directory, where the elliptic point height is currently an open task.

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